Reconstructing Sprint-Start Jira State for Leakage-Aware Outcome Prediction

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***Abstract*—A present-day Jira export shows how a sprint ended, not what the team knew when it began, so sprint-risk models trained on it can learn from hindsight. We make three contributions. First, a sprint-start reconstruction: for eight open-source TAWOS projects we replay the change log back to the first day of each sprint and rebuild the board as it stood (2,602 sprints, 1,691 with committed work), recording where missing history forces a static fallback. Second, a leakage-aware evaluation of planning-time classifiers. On a time-ordered split, scope-only logistic regression ranks spillover at ROC-AUC 0.885, but only 0.772 on the 110 test sprints whose scope has fully recorded history; gradient boosting ranks late closure at 0.751 and, trained on TAWOS alone, above chance in seven of eight testable outside projects from MariaDB and MongoDB (mean AUC 0.620). Two reproduced leakage mistakes inflate F1 by up to 0.24, shuffling sprints before splitting adds 0.07 to late-closure AUC, and dropping unfinished prior sprints from the history features lifts leave-one-project-out AUC for logistic regression from 0.542 to 0.629. Third, in-sprint warnings under the same protocol: progress features lift spillover AUC from 0.830 to 0.926 at 25% of the sprint and 0.957 at 50%, while late closure barely changes. The result is a reconstruction and an evaluation method, not a deployable risk tool: results vary across projects, and model-based what-if tables say nothing about cause.**

***Index Terms*—Scrum, sprint analytics, Jira, machine learning, explainability, Shapley values, data leakage**

I. Introduction

A sprint misses its boundary in two ways that Jira records by itself. It spills over when an issue committed at the start is still unresolved at the planned end, and it closes late when Jira stamps its completion more than a day past that end. Neither is a judgment on the team. We ask whether either can be ranked from what was known on the first day of the sprint, and whether watching the sprint unfold adds anything. To keep hindsight out, every feature is rebuilt only from events time-stamped at or before the point of prediction.

That line is harder to hold than it sounds, because a present-day Jira export is no snapshot of planning day. Issues move between sprints, estimates get revised, and several sprints are usually open at once, so the current issue-to-sprint link cannot tell work committed at planning from work added later. We replay the TAWOS change log back to each sprint's start and record, per project, where a missing history forces a static fallback. A second leak is easier to miss. In 71% of our sprints the previous sprint had not reached its planned end when the next began, so any summary of it can carry an outcome nobody could have known yet. We measure what excluding those unfinished sprints costs.

This is a data and evaluation contribution, not a deployable risk tool. It consists of a sprint-start reconstruction from the change log, with an account of how much of it had to fall back on static values; a leakage-aware evaluation of planning-time spillover and late-closure classifiers, with per-project time splits, leave-one-project-out (LOPO) testing, project-cluster bootstrap intervals, and explicit checks for zero-scope sprints, history provenance and two reproduced leakage mechanisms; and, under the same protocol, in-sprint warnings at 25%, 50%, 75% and 100% of the planned duration set against the planning-time models, plus numeric targets and exact Shapley explanations over feature groups. A sprint advisor packages the fitted models and is descriptive only. Fig. 1 lays out the pipeline and Table II pins each headline result to its model, split, population and comparator. All of it rests on eight open-source projects plus ten outside projects from MariaDB and MongoDB, so we read the numbers as evidence about those projects, not as a general benchmark.

II. Related Work

Machine learning on Agile data has mostly chased effort. Choetkiertikul et al. [1] estimated story points from issue text with deep learning; Porru et al. [2] did the same from issue report fields. When Tawosi et al. [3] replicated that work on TAWOS, they found simple baselines kept pace once the feature set was fixed, a pattern we meet again on spillover. Sprint2Vec [4] instead learns sprint embeddings from issue-level data to regress on how much is delivered and how good it is, while Hearty et al. [5] forecast velocity with a dynamic Bayesian network. All predict effort or volume delivered. None of them targets spillover or late closure as such, or reports rebuilding the sprint-start state from the change history; our search was not systematic, so this describes these studies, not the whole field. Table I puts them side by side.

Two sprint-level classification studies prompted our leakage checks. Pérez Castillo et al. [6] classified sprint performance from burndown features visible only while the sprint runs, which makes the task retrospective rather than a forecast. Obike et al. [7] reported near-perfect deliverability AUC after running SMOTE oversampling [8], but without describing a train-test split taken first. We did not read either implementation and

TABLE I
THE NEAREST PRIOR WORK AND WHAT THIS STUDY ADDS.

Study	Unit	Target	Sprint-start state
[1]–[3]	Issue	Story points	Not applicable
[4]	Sprint	Delivery and quality	Not reported
[5]	Iteration	Velocity	Not reported
[6]	Sprint	Performance class	In-sprint burndown
[7]	Sprint	Deliverability	Not reported
This work	Sprint	Spillover, late closure	Rebuilt (change log)



Fig. 1. The three-stage pipeline. Every stage uses the same reconstructed cohort, the same per-project time split and the same LOPO folds.

make no claim about their published numbers; instead we reproduce both mechanisms on our own data, to show what each would have cost us (Section IV-H).

For the explanations we take Shapley values from Lundberg and Lee [9], but compute them exactly, over feature groups rather than approximately over single features, since Jiarpakdee et al. [10] showed how correlated metrics distort a per-feature reading; Molnar [11] covers the general case. The nearest relative to our in-sprint setup is outcome-oriented predictive process monitoring [12] and its alarm-based extensions [13], in which a running case is scored at a sequence of prefixes. We did not find it applied to Scrum sprints with a reconstructed state.

III. DATA AND METHODS

A. Dataset and Sprint-Start Reconstruction

Our data comes from the Sprint, Issue and Change_Log tables of TAWOS [14]. We keep the eight open-source Jira projects with at least 30 sprint records apiece: Apache Mesos, Appcelerator Studio, Atlassian Confluence Cloud and Server, LSST Data Management, Mule, Titanium SDK and Titanium Mobile. The export carries 2,657 sprint rows, 145,309 issues and 88,364 change events on the Sprint and Story Points fields. A few sprints are excluded: the 44 still marked FUTURE or ACTIVE, and three CLOSED rows that finish before they begin. Since the history features need a predecessor, we also drop each project's first eligible sprint as a cold start. That leaves 2,602 modelled sprints, 911 (35.0%) committing nothing and 1,691 carrying real scope.

We rebuild the committed scope rather than read it off the final export. Starting from the latest recorded state, the code walks each issue's Sprint and Story Points changes back to the sprint-start timestamp; an issue counts as committed only if, at that moment, it existed, belonged to the sprint, and had not yet been resolved. That gives 20,603 issue-sprint commitments. When an issue has no recorded change for a field, we fall back on its static value and log it: 7.2% of commitments lean on the static sprint link and 45.1% on static story points, and 73.5% of issues carry no story points at all and count as zero. Coverage is uneven, from 54.1% of commitments with observed sprint history in Confluence Cloud up to 99.1% in Confluence Server, so Section IV-B re-runs the analysis on provenance-restricted subsets.

B. Outcomes

Late closure is 1 when Jira's completion timestamp runs more than a day past the planned end, and is defined for all 2,602 sprints. Spillover is 1 when at least one committed issue is still unresolved at the planned end; we score it only on the 1,691 nonempty sprints, since a sprint that commits nothing cannot spill and would be a free correct answer. Three numeric targets sit alongside the labels: issues completed and carry share (the fraction of committed issues left unfinished) on the nonempty sprints, and days late, capped at 30, on all of them. Table II gives each task's population and denominator beside its headline result.

C. Features and the Parallel-Sprint Timing Fix

Every feature is read off the reconstructed sprint-start state. Scope is the committed issue count and story points; sprint length is the planned duration in days; recent history rolls up the project's earlier sprints (mean delivered points and mean completion ratio over the last three, the last carryover, and how many such sprints exist); project identity is one-hot encoded. The strict set adds the number of other sprints open at the start (mean 1.9, as many as 11), team capacity (mean issues completed and committed over recent sprints, and the new scope against that capacity), and past lateness (mean days late and the share of late sprints). The two sets differ in the meaning of "recent". The legacy set, from our first study, summarises the immediately preceding sprints whether or not they had finished. The strict set admits only sprints whose completion falls before the current start. We run both; results use the strict set unless marked legacy.

D. Models and Validation

On the labels we line up an always-positive rule, a project base-rate predictor, a scope-only logistic regression (committed issues, story points, sprint length), logistic regression over all features with balanced class weights, a random forest (300

TABLE II

STUDY DESIGN AND HEADLINE RESULTS. n (POSITIVES) IS THE TEST DENOMINATOR OF EACH FIGURE; CHECKPOINT ROWS COUNT SPRINTS STILL OPEN. DIFFERENCES ARE ROW MODEL MINUS COMPARATOR, WITH 95% PROJECT-CLUSTER BOOTSTRAP INTERVALS; ABSOLUTE INTERVALS WHERE THE ARTIFACT REPORTS THEM. GB: GRADIENT BOOSTING; RF: RANDOM FOREST; LR: LOGISTIC REGRESSION; MAE: MEAN ABSOLUTE ERROR; SD: STANDARD DEVIATION; SE: STANDARD ERROR; SCOPE ONLY: COMMITTED ISSUES, STORY POINTS AND SPRINT LENGTH; STRICT AND LEGACY NAME THE HISTORY FEATURE SET. ROW 16 TRAINS ON TAWOS AND TESTS ON MARIADB WITHOUT STORY POINTS (SP), WITH A SPRINT-LEVEL INTERVAL; ROW 17 IS MEAN $\pm$ SD OVER FIVE TIME-ORDERED REPEATS; ROW 18 IS MEAN $\pm$ SD ACROSS THE EIGHT OUTSIDE PROJECTS WHERE LATE CLOSURE CAN BE MEASURED.

	Task	Split	n (pos.)	Model, features	Result	Difference [95% CI]	Comparator
1	Late closure	Time	786 (246)	GB, strict	AUC .751	+.111 [.028, .258]	Project rate .640
2	Late closure	Time	786 (246)	GB, strict	AUC .751	+.221 [.098, .369]	Scope-only LR .530
3	Late closure	LOPO	2,602 (1,006)	RF, strict	AUC .642	+.110 [.018, .215]	Scope-only LR .532
4	Late closure	LOPO	2,602 (1,006)	RF, legacy	AUC .600	—	Per project .531–.780
5	Spillover, nonempty	Time	511 (482)	LR, scope only	AUC .885 [.818, .922]	+.073 [.007, .117]	GB, strict .812
6	Spillover, nonempty	LOPO	1,691 (1,575)	LR, scope only	AUC .837	−.017 [−.051, .007]	LR, strict .854
7	Spillover, nonempty	LOPO	1,691 (1,575)	RF, legacy	AUC .815	—	Per project .706–.950
8	Spillover, all sprints	Time	786 (453)	RF, legacy	AUC .986	—	Nonempty .852 [.774, .911]
9	Spillover, full history	Time	110 (94)	LR, scope only	AUC .772, SE $\approx$ .05	—	Row 5
10	Spillover, 25%	Time	507 (478)	GB, strict + progress	AUC .926	+.097 [.054, .163]	No progress .830
11	Spillover, 50%	Time	501 (472)	GB, strict + progress	AUC .957	+.123 [.068, .188]	No progress .835
12	Late closure, 50%	Time	775 (246)	GB, strict + progress	AUC .760	+.008 [−.008, .031]	No progress .751
13	Issues completed	Time	511	GB (abs. error)	MAE 1.96	−.13 [−.31, .00]	Scope $\times$ rate 2.10
14	Carry share	Time	511	GB (abs. error)	MAE .201	−.031 [−.050, −.012]	Train median .232
15	Days late	Time	786	GB (abs. error)	MAE 2.48	−.18 [−.34, .03]	Train median 2.66
16	Late closure	MariaDB	391 (106)	GB, strict, no SP	AUC .710	+.134 [.034, .223]	Scope-only LR .576
17	Late closure	Time $\times$5	$\approx$520 (177)	GB, strict	AUC .696 $\pm$ .054	—	Shuffled 70/30 .765 $\pm$ .017
18	Late closure	10 outside	8 projects	GB, strict, no SP	Mean AUC .620 $\pm$ .090	—	Scope-only LR .362

trees, depth 5, minimum leaf 3, balanced weights), and gradient-boosted trees: scikit-learn [15] histogram gradient boosting at 200 iterations, depth 3 and a 0.05 learning rate (our first study used XGBoost [16]; all boosting figures here are scikit-learn). The numeric targets get ridge regression, a random forest [17], and gradient boosting under squared-error and absolute-error loss, each against train-median, prior-value and scope-times-completion-rate baselines. We ran no hyperparameter search and fixed the 0.5 threshold for precision, recall and F1 before seeing a result. ROC-AUC is our headline measure, since an always-positive rule already posts F1 0.971 on spillover, with the Brier score (mean squared error of the probabilities) beside it.

For the main test we sort each project's sprints by time and train on the earliest 70%, leaving 1,816 training and 786 test sprints for late closure and 1,180 and 511 for spillover. We also repeat it five times in time order, training on each project's first 40 to 80% and testing on the next 20%, and, for contrast, run five shuffled 70/30 splits and five shuffled five-fold cross-validations. LOPO holds out a whole project and trains on the other seven, with the held-out project's history still drawn from its own past (Section IV-C). For uncertainty we resample the eight projects as whole units 2,000 times and report 95% percentile intervals of the metric or a paired difference, since sprints within a project are hardly independent. The seed is 42 throughout.

E. Explanations, Advisor and Checkpoints

We compute the Shapley values exactly, over the seven feature groups, in their interventional form: for a given sprint and each of the 128 group coalitions, the coalition's features are pinned to that sprint's values while the rest are drawn from a background sample of 40 training sprints, and the coalition is scored by the model's mean prediction over that sample. The attributions then sum exactly to the prediction minus the background mean, and for each group we report its share of the mean absolute attribution over 150 random test sprints (100 at the in-sprint checkpoint). The advisor wraps the fitted models: for a planned sprint it returns both probabilities, the expected issues completed with an 80% interval from quantile boosters, the median and 90th-percentile days late, and a what-if table across alternative scopes. A what-if table is only a model's expectation, so we also set the heaviest quartile of scope-relative-to-capacity against the lightest, on realised carry share and lateness, with cluster bootstrap intervals.

For the in-sprint work we rebuild the sprint state at 25%, 50%, 75% and 100% of the planned duration, from events time-stamped at or before each point: committed issues done and remaining, points left, the gap between work finished and time elapsed, the observed and required completion rates, and issues or points added or dropped since the start. A sprint already closed by a checkpoint is dropped from it, its outcome no longer in doubt (of the 2,602 sprints, 98.6% are still open at 25%, 96.2% at 75% and 65.5% at 100%), and we do not run spillover at 100% because the state settles it. As before, each progress-aware model is set against the same algorithm on planning-time features alone and against a learning-free rule on the same split (one minus the share done for spillover, committed issues still open for late closure).

IV. RESULTS

A. Planning-Time Classification

Table III collects the main time-split results under strict features. Every spillover figure here is on nonempty sprints, and Section IV-B adds the stricter provenance check. Late closure is rankable, if only moderately: gradient boosting reaches ROC-AUC 0.751, the random forest 0.715 and logistic regression 0.680, while the scope-only model stays at 0.530, close to chance. Only gradient boosting's paired cluster-bootstrap interval clears the project base rate of 0.640 (Table II, rows

TABLE III
PLANNING-TIME RESULTS, STRICT FEATURES, TIME-ORDERED SPLIT (FIXED 0.5 THRESHOLD).

Task	Model	n	F1	AUC	Brier
Late	Always positive	786	.477	.500	—
Late	Scope-only LR	786	.336	.530	.244
Late	Project base rate	786	.470	.640	.221
Late	Logistic regr.	786	.495	.680	.224
Late	Random forest	786	.502	.715	.207
Late	Gradient boosting	786	.553	.751	.176
Spill.	Always positive	511	.971	.500	—
Spill.	Scope-only LR	511	.696	.885	.230
Spill.	Logistic regr.	511	.842	.827	.161
Spill.	Random forest	511	.880	.840	.115
Spill.	Gradient boosting	511	.974	.812	.049

1 and 2). Spillover is the reverse. The scope-only model hits 0.885 (cluster interval 0.818 to 0.922) and everything richer trails it: logistic regression 0.827, gradient boosting 0.812 (row 5), and the random forest 0.840, whose paired difference of -0.045 has an interval of -0.088 to 0.010. That 0.885 leans partly on static fallbacks, whose present-day values can carry hindsight; Section IV-B rescores without them.

Population matters as much. With legacy features the spillover forest scores 0.986 AUC on all 2,602 sprints and 0.852 on the 1,691 nonempty ones. That gap is the 35% of sprints negative by construction, so a study that keeps them looks roughly 0.13 AUC stronger than it is. Nor does the late-closure headline ride on one cut: over five time-ordered repeats gradient boosting averages 0.696 ± 0.054 (row 17), rising from 0.62 with 40% of each history for training to 0.75 with 70%; spillover averages 0.806 ± 0.028.

B. Timing and Provenance

Swapping legacy history features for strict ones lifts late-closure AUC for most models under both protocols, and the lift is biggest across projects. Under LOPO, logistic regression climbs from 0.542 to 0.629 and the random forest from 0.600 to 0.642; on the time split, logistic regression goes 0.636 to 0.680 and gradient boosting 0.726 to 0.751. Spillover barely moves (about 0.02 either way, inside the noise), since scope is identical in both sets. No headline flips, but the legacy features were never known at planning time.

Provenance is the bigger caveat, and the main spillover estimate should be read with it. We rerun the scope-only model and the legacy forest on two nested subsets. Require that every committed item have at least one Sprint-field change recorded before the sprint start, so that its membership is observed rather than assumed, and 1,174 sprints remain (356 in the test period, 335 positive): the scope-only model scores 0.875 and the forest 0.850. Require a recorded Story Points change as well, and 358 remain (110 in the test period, 94 positive and 16 negative): 0.772 and 0.652 (row 9). The artifact carries no bootstrap for these subsets, so we give the analytic Hanley–McNeil standard error instead: 0.054 for the smaller one, an approximate 95% range of 0.67 to 0.88 that overlaps the main estimate's 0.818 to 0.922. With 16 negatives, 0.77 is no replacement estimate; it shows only that the 0.885 is not yet established on fallback-free data.

C. Cross-Project Generalisation

LOPO keeps each held-out project's own earlier sprints as history inputs, so it asks whether a rule learned on seven projects carries to an eighth that has a track record, not whether it survives a cold start. Within that test the project, far more than the model, moves performance. With legacy features the late-closure forest pools to 0.600 but runs from 0.531 on Titanium Mobile (48 late sprints out of 212) up to 0.780 on Appcelerator Studio; the nonempty-sprint spillover forest from 0.706 on Confluence Cloud, whose 197 nonempty sprints are hard to tell apart, up to 0.950 on Appcelerator Studio. Under LOPO the plain always-late rule (F1 0.558) beats every learned model on F1: a threshold tuned elsewhere does not travel. Concept drift [18] explains part of this: under our split, Titanium SDK's late-closure rate drops from 71% in training to 33% in test, and Apache Mesos's from 38% to 9%.

D. Outside Checks on MariaDB and MongoDB

LOPO never leaves TAWOS, so we ran the frozen pipeline on two MariaDB projects it had never seen, MaxScale and ColumnStore: 391 sprints from 2016 to 2026, 106 of them late, converted to the TAWOS tables with no change to reconstruction, labels, features or models. MariaDB records no story points, so these runs drop the point features on both sides (the TAWOS late-closure AUC stays at 0.751 without them). Trained on TAWOS alone, with no project identity, gradient boosting ranks the pooled MariaDB late closures at 0.710 (sprint-bootstrap interval 0.637 to 0.773), against 0.576 for scope only (row 16).

We then added eight MongoDB projects (Compass, WiredTiger, Shell, Database Tools and the Java, C, C++ and PHP drivers; 526 sprints) from its public Jira, and scored all ten outside projects one at a time, with the decision threshold fixed on TAWOS (Youden's J on out-of-fold training predictions). Late closure can be measured in eight of them; the other two have only three on-time sprints each, under our minimum of five per class. Mean AUC is 0.620 ± 0.090, above chance in seven and above scope only in all eight (row 18); the scope-only baseline averages 0.362, so on these teams committed scope ranks late closure the wrong way. Per project it is lower than the pooled MariaDB figure, which also rewards telling projects apart, and intervals are wide at 20 to 40 sprints apiece. Sensitivity runs from 0.11 to 0.86 because late rates run from 7% to 59%: the ranking travels, the threshold does not. Spillover can be measured in only four projects (mean AUC 0.734, scope only 0.742); in the rest nearly every sprint spills.

E. What Explains Each Outcome

Fig. 2 lays out the exact Shapley shares for the two planning-time random forests. For late closure, past lateness takes 38% of the attribution and project identity 22%, with sprint length and capacity at 13% and 12%, and committed scope 9%. For spillover, committed scope takes 41% and capacity 37%: how much was taken on against how much the team usually clears. A group ablation on the legacy forest agrees: pull scope out and nonempty spillover AUC collapses from 0.852 to 0.652,

TABLE IV
IN-SPRINT ROC-AUC BY CHECKPOINT: GRADIENT BOOSTING WITH AND WITHOUT PROGRESS FEATURES, AND THE LEARNING-FREE PROGRESS RULE. EACH CHECKPOINT SCORES ONLY SPRINTS STILL OPEN AT THAT POINT, SO THE PLANNING-ONLY FIGURES SHIFT AT 100%; THE LATE-CLOSURE PLANNING-ONLY FIGURES AT 25% TO 75% DIFFER ONLY IN THE FOURTH DECIMAL.

Task	Model	Split	25%	50%	75%	100%
Spill.	Planning only	Time	.830	.835	.827	—
Spill.	With progress	Time	.926	.957	.953	—
Spill.	Progress rule	Time	.667	.767	.802	—
Spill.	Planning only	LOPO	.801	.815	.804	—
Spill.	With progress	LOPO	.860	.897	.927	—
Spill.	Progress rule	LOPO	.610	.715	.827	—
Late	Planning only	Time	.751	.751	.751	.717
Late	With progress	Time	.754	.760	.749	.704
Late	Progress rule	Time	.479	.483	.486	.484
Late	Planning only	LOPO	.645	.652	.647	.672
Late	With progress	LOPO	.658	.667	.664	.657
Late	Progress rule	LOPO	.518	.517	.521	.510

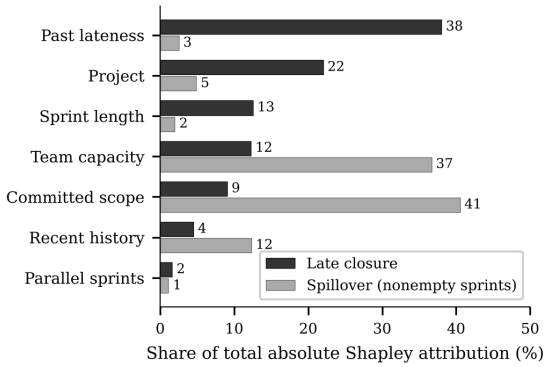


Fig. 2. Share of mean absolute exact Shapley attribution by feature group for the two random-forest classifiers (strict features), over 150 time-split test sprints.

while late closure barely moves, 0.705 to 0.690. The legacy forests' reliability bins (the late-closure 0.7–0.8 bin runs 35% late) warn against reading their scores as probabilities, so we use them only as rankings.

F. Numeric Targets and the Advisor

Issues completed is mostly arithmetic. Multiply committed issues by the team's recent completion rate and the mean absolute error is 2.10 issues (R^2 0.42); the best learned model, a booster trained on absolute error, manages 1.96 (R^2 0.46), a difference whose interval grazes zero (row 13). Carry share is all but unpredictable (R^2 near zero for every model), though that booster does trim the error from 0.232 to 0.201 (row 14). Days late gains little: the absolute-error booster reaches 2.48 against the training median's 2.66. The advisor's 80% intervals for issues completed cover 84% of outcomes on the time split and 85% across projects, with a mean width of 5.4 issues against the scope-times-rate baseline's 8.4, though coverage sags to 70% for sprints of 13 or more issues.

The what-if table is not advice. Sprints planned heaviest relative to capacity carry over just 3.2 percentage points more of their scope than the lightest quartile (interval -2.3 to $+8.4$) and close late 5.6 points more often (-5.2 to $+16.4$).

G. In-Sprint Early Warning

Table IV shows what progress buys. For spillover it buys a great deal, and early: on the time split gradient boosting moves from 0.830 on planning features to 0.926 at the 25% checkpoint, 0.957 at 50% and 0.953 at 75%, gains of +0.10, +0.12 and +0.13 over the same model without progress, each with an interval clear of zero (0.054 to 0.163 at 25%). Across projects the gains are +0.06, +0.08 and +0.12, with intervals clear of zero for gradient boosting, though at 25% the logistic and forest gains do not. The learning-free rule reaches 0.667, 0.767 and 0.802 on the time split, so the model is worth between 0.15 and 0.26 AUC over a bare count of open issues. For carry share, the rule "one minus the share done so far" (0.165, 0.115, 0.064 mean absolute error) matches the model with progress (0.146, 0.112, 0.070) from halfway on.

Late closure, by contrast, barely moves. Gradient boosting scores 0.751 on planning features and 0.754, 0.760, 0.749 and 0.704 with progress at the four checkpoints. Of 24 paired comparisons with the same model without progress, 22 intervals include zero and the other two gains are under 0.03 AUC; on MariaDB the mid-sprint gain is +0.019 (interval -0.050 to 0.089). The open-count rule hovers near 0.5. At 50% the explanations agree: progress takes 34% of the spillover attribution but only 6% of the late-closure one, where past lateness still holds 37%.

H. Leakage Case Studies

We reproduce two mistakes on our own data with our first study's legacy logistic regression, scoring F1 at the fixed threshold before and after each. Feed in the sprint's own completion ratio, known only once the sprint is over, and spillover F1 jumps from 0.756 to 0.991 and AUC from 0.834 to 1.000, because the feature just restates the label; late closure hardly moves (+0.010), since finishing the work is not what decides when somebody clicks "complete sprint". Run SMOTE over the whole pool before splitting and late-closure F1 inflates from 0.457 to 0.646, as synthetic copies of test sprints leak into training, while spillover is untouched: at 94% positives there is next to nothing to oversample. A third mistake is shuffling sprints before splitting: over five repeats, shuffled 70/30 splits

give late closure 0.765 ± 0.017 against 0.696 ± 0.054 in time order, since it trains on a team's later sprints; spillover hardly moves (0.813 against 0.806).

V. Threats to Validity

Construct. Jira's completion timestamp marks when a sprint was closed, not when its work was done, so late closure is an administrative event. Spillover says only that at least one committed item went unresolved.

Reconstruction. Replaying the change log beats trusting a static export, but what was never recorded cannot be recovered, and Section IV-B shows how the story-point fallback caps the spillover estimate. The strict history features close the parallel-sprint leak but cannot promise that the rebuilt state is what the team saw. We read completion from each issue's final resolution date, so an issue reopened and resolved again counts at that later resolution, which can only make early progress look slower.

Evaluation. Eight projects make for wide cluster intervals, and our bootstrap over projects is exploratory, not confirmatory. We ran no decision-utility analysis. Prevalence drives the thresholded metrics, and the drift within projects means any deployed model would need local recalibration. The Shapley values describe the fitted models, not causes.

External. Every project here is open-source, and the outside checks add two sources and ten projects, with spillover testable in only four of them. Enterprise teams that close sprints on schedule may behave differently.

VI. Conclusion

This study contributes a protocol more than a tool, and under it reconstructed Jira history does carry signal, real but specific to each outcome. Spillover is a work outcome: committed scope accounts for most of it at planning time (scope-only AUC 0.885 on nonempty sprints, time split), gradient boosting with progress reaches 0.96 at mid-sprint, and at 94% prevalence the question worth asking is which sprints carry the most unfinished work, not whether any will. Late closure is an administrative-timing outcome: planning-time features reach 0.751 with history held to sprints that had genuinely finished (0.726 with the legacy set), past lateness and project identity take 60% of the forest's attribution, and progress adds little. Performance varies widely by project, so the exact numbers should not be expected to carry over, though late closure still ranked above chance in seven of eight testable outside projects. The lessons are about data and evaluation: rebuild the sprint-start state, drop the sprints that commit nothing, honour parallel-sprint timing, audit for leakage one label at a time, since the same mistake costs 0.24 F1 on one outcome and 0.01 on the other, and keep time order when splitting.

For a team this gives two narrow things. At planning, the scores rank sprints by risk, so a team knows which ones to watch; they are rankings, not calibrated odds, and any alert threshold has to be set locally. During the sprint, the 25% and 50% checkpoints flag likely spillover earlier and better than counting open issues. We have not shown that acting on a score improves delivery, or that cutting scope causes a lower carry share; that needs a prospective or controlled study.

Every result runs off seed 42 under Python 3.13, scikit-learn 1.8 (1.9 for the outside checks and repeated splits), pandas 2.2 and XGBoost 3.1. The artifact at https://github.com/hani-ziad/SwEng_202610 holds the data export, the code, 50 automated tests and every result file behind these numbers.

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